
MA 656: Stochastic Calculus for Financial Engineering

Course Code	MA 656
Course Name	Stochastic Calculus for Financial Engineering
L-T-P-C	2-1-0-3
Prerequisites	Good knowledge of multivariable calculus, probability, statistics and stochastic process. Some knowledge of MATLAB/R/Spread sheets Packages.
Intended for	M.S./Ph.D.
Distribution	Elective
Approval	6th Senate (May 2014)

Course Contents

- **Module I**
Definitions and introduction to Financial instruments and derivatives, no arbitrage principle, risk-neutral probability measure. [6 Lectures]
- **Module II**
Give 1-2 lectures on Stochastic process, distribution functions, (again because students may need to recall it for building comfort level), Random walk, Brownian and Geometric Brownian Motion, Lévy's construction. Reflection principle, hitting times, scaling properties. Theory of Martingales, filtrations, adapted processes, Optional Sampling Theorem. [7 Lectures]
- **Module III**
Quadratic variation and Brownian motion, Itô integral, properties of stochastic integral, Representation Theorem, Lévy's characterisation of Brownian motion, Girsanov's Theorem, Feymann Kac Theorem, Ito's Formula . [8 Lectures]
- **Module IV**
Self-financing strategies, martingale measures, risk-neutral pricing. Applications of Stochastic Calculus in Option Pricing. Black-Scholes Model and Pricing Formula, European options, Risk management strategies for options. [7 Lectures]

Text Books

1. D Duffie, **Dynamic Asset Pricing Theor.**, Princeton 1996.
2. G Grimmett & D Stirzaker, **Probability and Random processes**, Oxford 1982.
3. Thomas Mikosch, **Elementary Stochastic Calculus with Finance in view**, World Scientific, 2006.
4. S. E. Shreve, **Stochastic Calculus for Finance**, Vol. I & Vol. II, Springer, 2004.

References

1. Ralf Korn, Elke Korn, **Option Pricing and Portfolio Optimization**, American Mathematical Society, 2000.
2. M. Capinski and T. Zastawniak, **Mathematics for Finance: An Introduction to Financial Engineering**, Springer, 2005.
3. N. H. Bingham and R. Kiesel, **Risk Neutral Valuation**, 2nd Edition, Springer, 2004.